

INTER-IIT TECH MEET

INTERNAL HACKATHON

MULTI-MODEL SELECTION - PROBLEM STATEMENTS

SUBMISSION STRUCTURE - MANDATORY

Task 01 and Task 02 are separate tasks. Participants must create **two separate repositories** and **two separate PPT presentations** - one repository and one PPT for each task. Do not combine the two tasks into a single repo or a single presentation.

Overview

The internal hackathon consists of two independent selection tasks. Participants are expected to demonstrate both technical implementation skills and the ability to optimize, benchmark, and clearly communicate their design decisions.

Requirement	Details
Task 01	Image Captioning on the open-source VRSBench dataset
Task 02	Text-Based Object Removal & Replacement under CPU-only constraints
Repositories	Two separate repos - one dedicated to each task
Presentations	Two separate PPTs - one dedicated to each task

Participants may work on both tasks independently. The implementation, experiments, observations, and presentation for each task should be self-contained.

TASK 01

Image Captioning on VRSBench

Open-source dataset based image captioning with an emphasis on training-time optimization and experimental observations.

1. Problem Statement

Perform **image captioning** using the open-source **VRSBench** dataset. The goal is to build and train an image-captioning system and investigate how optimization techniques affect training efficiency and model performance.

The task is intentionally focused not only on obtaining a working captioning model, but also on understanding and documenting the optimization decisions made during training.

2. Objective

- Train an image-captioning model on the VRSBench dataset.
- Experiment with practical optimization techniques during model training.
- Observe how the selected optimizations affect training behavior, convergence, resource usage, and captioning quality.
- Document the reasoning, observations, and measurable outcomes of the experiments.

3. Evaluation Focus

Evaluation is centered on optimization and analysis.

The evaluation will consider the **optimization techniques implemented** and the **observations documented during model training**. Participants should therefore clearly explain what was changed, why it was changed, and what effect was observed.

4. Expected Work

Requirement	Details
Dataset	Open-source VRSBench dataset.
Core task	Generate natural-language captions for input images.
Optimization	Participants are encouraged to investigate and implement meaningful training optimizations.
Analysis	Record training observations, comparisons, and relevant metrics or resource measurements.
Reproducibility	The task implementation should be sufficiently documented so that experiments can be reproduced from the repository.

5. Suggested Experimentation Areas

- Training configuration and learning-rate strategies.
- Batching, gradient accumulation, mixed precision, or other memory/time optimizations where supported.
- Data pipeline and preprocessing efficiency.
- Model-side efficiency measures and inference considerations.
- Comparison of baseline versus optimized training runs, with observations supported by measured results.

Task 01 Submission

Submit this task through its own repository and its own PPT. The repository/PPT should contain only the material relevant to the VRSBench image-captioning task.

TASK 02 Text-Based Object Removal & Replacement

CPU-only image editing task focused on compact models and edge-computing trade-offs.

Task Intent

Recreate a scaled-down version of a **text-driven Photoshop-style editing workflow**. The emphasis is on how effectively a participant can design a lightweight, CPU-feasible solution under practical edge-device constraints.

1. Problem Statement

Build a notebook that performs **text-based object removal and replacement** on input images. A user should be able to describe an object in natural language and request either its removal or replacement with a text-defined target.

The implementation should explore both **diffusion and non-diffusion approaches** wherever practical, while keeping the deployment footprint as small as possible.

The challenge is intentionally open-ended: participants are expected to investigate multiple techniques, make engineering trade-offs, and justify why their chosen approach is appropriate for CPU-based edge inference. Solution quality will therefore be considered together with model compactness and inference efficiency.

2. Objective

- Develop a text-driven image editing pipeline for object removal and object replacement.
- Explore multiple solution families, including non-diffusion methods and diffusion-based methods where they are feasible under CPU constraints.
- Minimize total model size and runtime dependencies without sacrificing the quality of the edited output.
- Demonstrate sound edge-computing reasoning: model selection, optimization, preprocessing/postprocessing, and practical deployment considerations.

3. Constraints

Requirement	Details
Inference hardware	CPU inference only. GPU-based inference must not be required for the submitted solution.
Model footprint	Use the smallest practical model(s) possible. Compactness should be treated as a core design objective, not an afterthought.
Task interface	The workflow should be text-driven: object identification and replacement intent should be expressed through basic text prompts/instructions.
Reproducibility	The notebook and repository must contain sufficient instructions, dependencies, and setup details to reproduce the reported results.
No fixed architecture	Participants may choose, combine, compress, distill, or otherwise optimize models/techniques, subject to the above constraints.

4. Deliverables

Requirement	Details
Working notebook	A runnable notebook demonstrating text-based object removal and replacement on representative input images.

Requirement	Details
Source repository	Complete codebase/repository containing the notebook, model/configuration files required to run it, dependency specification, and clear setup instructions.
Solution approach document	A concise technical document explaining the explored approaches, final architecture/pipeline, model choices, optimization decisions, and key trade-offs.
Benchmark report	Measured CPU inference time and relevant model-size information, along with a brief discussion of output quality and observed bottlenecks.

5. Selection Focus

Make the engineering trade-off explicit.

The submission should clearly document the relationship between **output quality**, **model footprint**, and **CPU inference time**, together with the reasoning behind the final approach.

6. Recommended Benchmarking Evidence

- CPU inference latency on a clearly specified hardware/software setup.
- Model size / parameter count and, where relevant, additional runtime dependency footprint.
- Representative examples of successful removals and replacements.
- Observed failure cases, bottlenecks, and trade-offs between quality and efficiency.

Task 02 Submission

Submit this task through its own repository and its own PPT. The repository/PPT should contain only the material relevant to the text-based image-editing task.

FINAL SUBMISSION CHECKLIST

The following structure is mandatory for the internal hackathon selection:

Item	Task 01 - VRSBench	Task 02 - Object Editing
Repository	Separate Repo 01	Separate Repo 02
PPT	Separate PPT 01	Separate PPT 02
Core output	Image captioning model + training experiments	Text-based removal/replacement pipeline
Primary emphasis	Optimization techniques + training observations	Quality + model compactness + CPU efficiency

IMPORTANT

Do not merge the tasks. There must be exactly **2 independent repositories** and **2 independent PPT presentations** for selection and evaluation.

End of Problem Statements